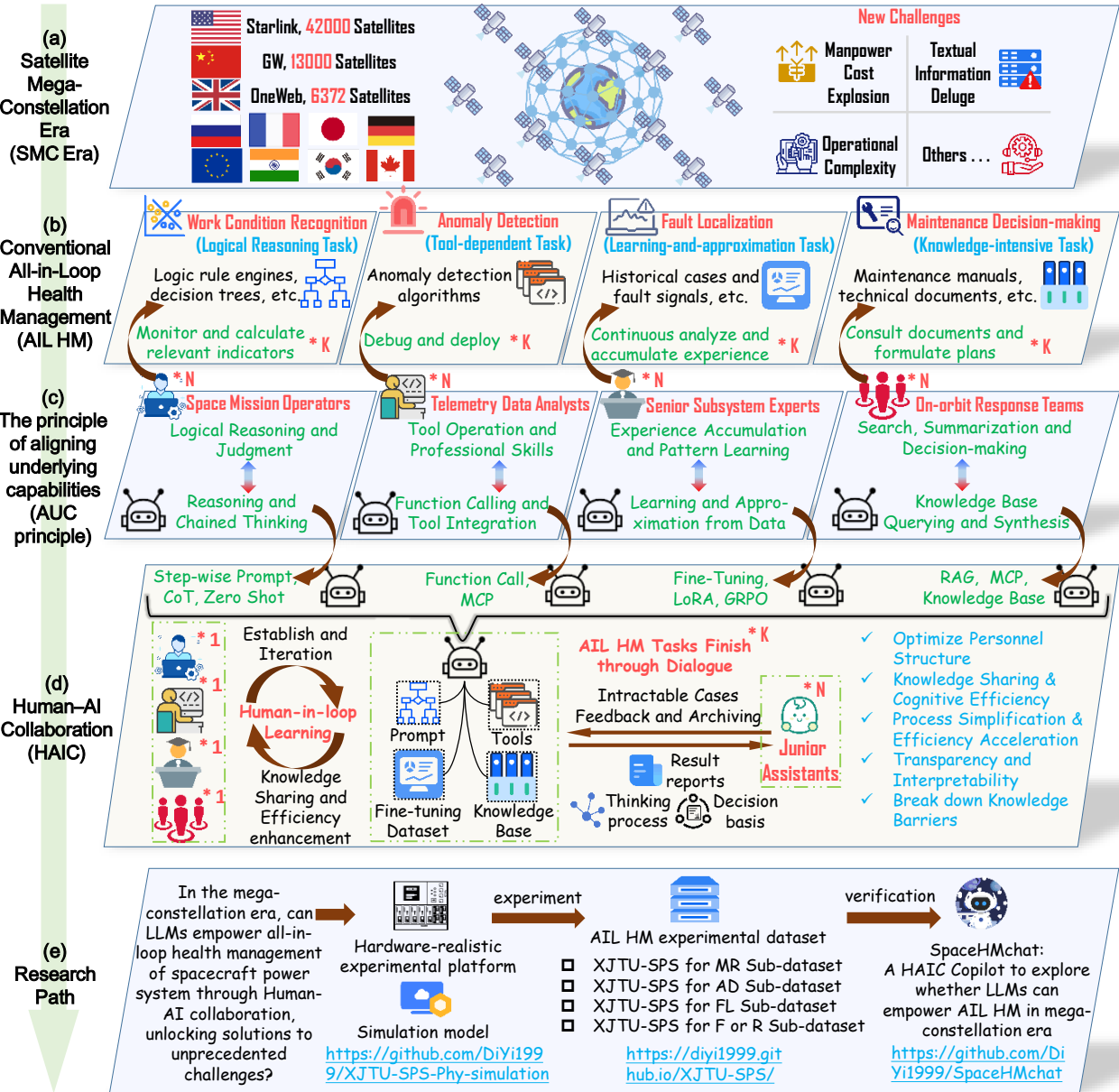




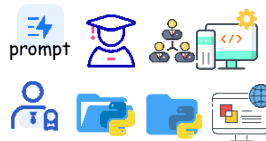
<https://huggingface.co/datasets/DiYi1999/XJTU-SPS>

SpaceHMchat

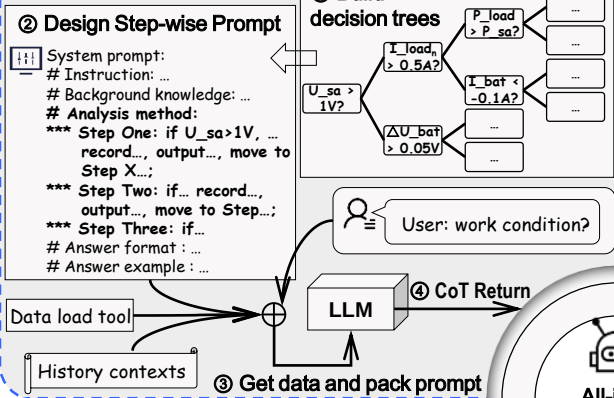
Soft and hardware support

unprecedented challenges in the mega-constellation era !

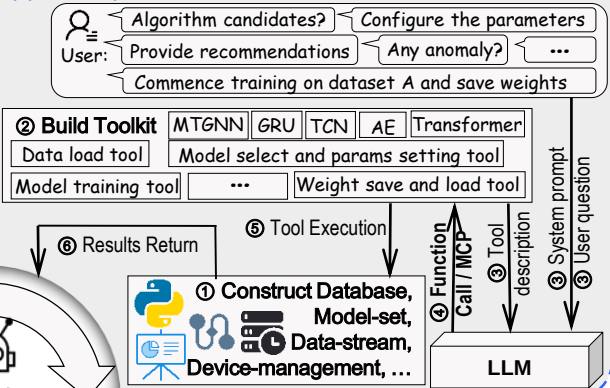
This work explores whether large language models can empower spacecraft health management in the mega-constellation era by establishing an effective Human-AI collaboration paradigm.



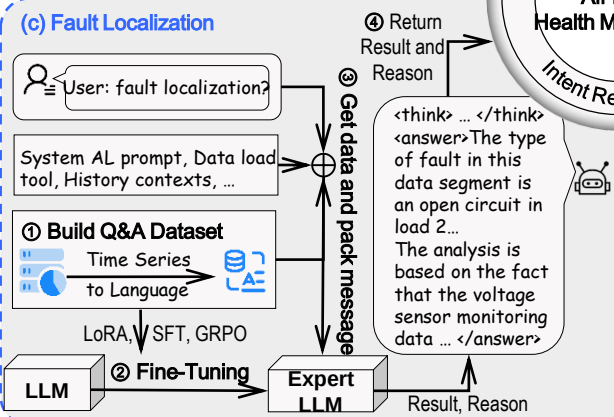
## (a) Work Condition Recognition



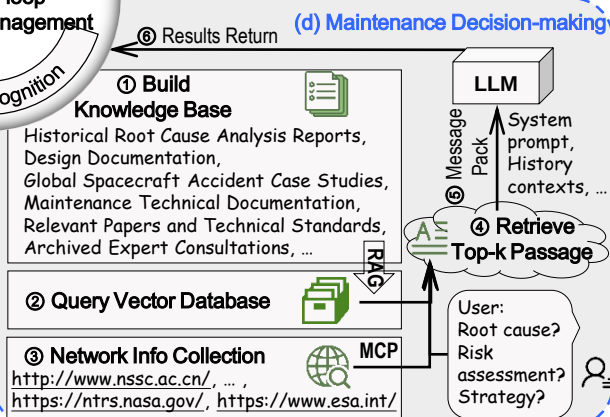
## (b) Anomaly Detection

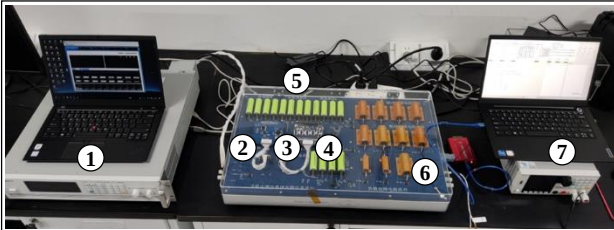


## (c) Fault Localization

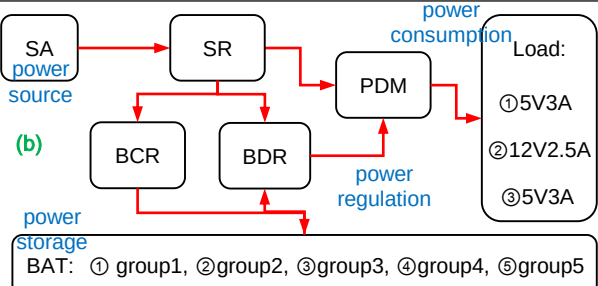


## (d) Maintenance Decision-making





(a) 1.Solar Array (SA) Simulator, 2.Shunt Regulator (SR),  
3.Battery Charge Regulator (BCR), 4.Battery Discharge Regulator (BDR),  
5.Battery Set (BAT), 6.Power Distribute Module (PDM), 7.Load Simulator.

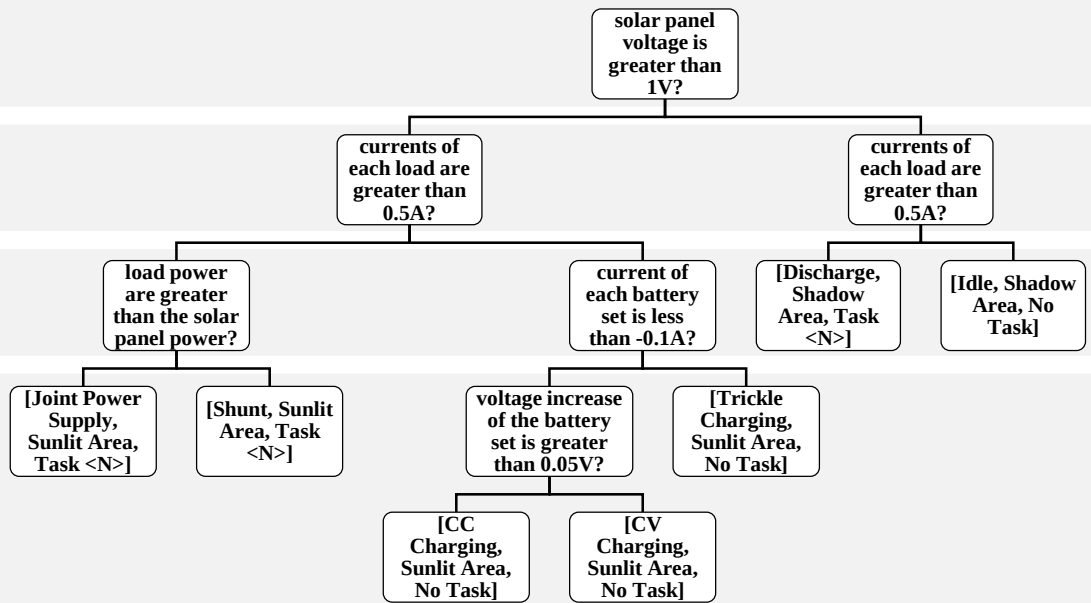


(c) XJTU-SPS Dataset

33 channel sensors: voltage, current, power and temperature sensors from SA, BCR, BATs, PDM, Loads, and Bus.

95 minute orbital period: simulate a low earth orbit spacecraft, covering various scenarios in sunlit and shadow areas.

| XJTU-SPS for F or R Sub-dataset (for Forecasting or Reconstruction Task)                                                | XJTU-SPS for MR Sub-dataset (for Work Mode Recognition Task)                                               | XJTU-SPS for AD Sub-dataset (for Anomaly Detection Task)                                                 | XJTU-SPS for FL or FD Sub-dataset (for Fault Localization / Fault Diagnosis Task)                                       |
|-------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Simulate the data of 4, 18, 24, 34, 90, and 94 orbits around the earth, respectively, with a sampling frequency of 1Hz. | Simulate working conditions including Fast Charge, Shunt, Trickle Charge, Joint, Idle, and Discharge, etc. | Simulate the situations when various anomalies occur during the operation, exceeding 700,000 timestamps. | Simulate 17 types of fault scenarios, including BAT open circuit, BCR short circuit, and Bus insulation breakdown, etc. |

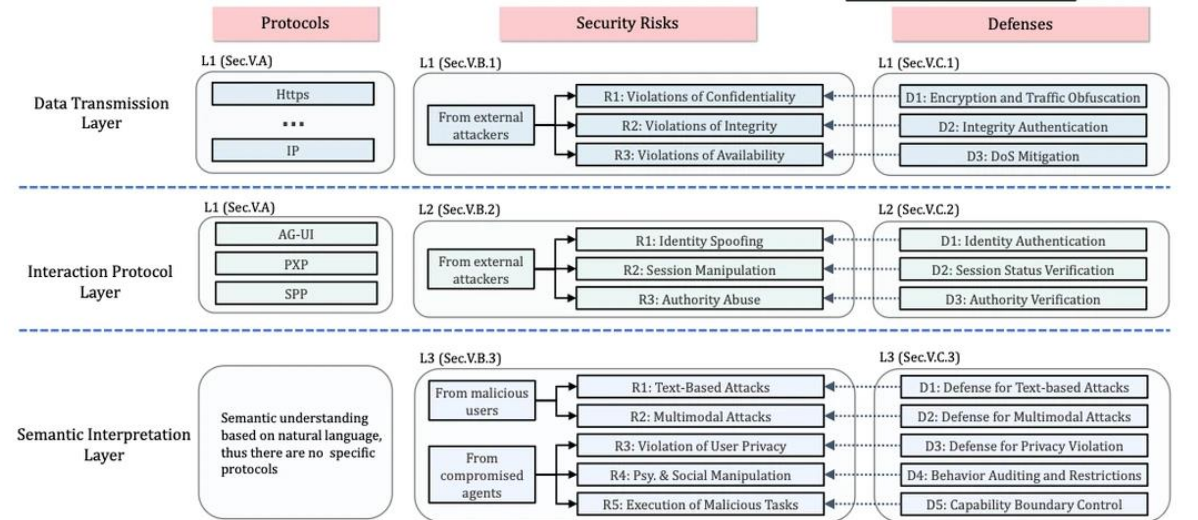
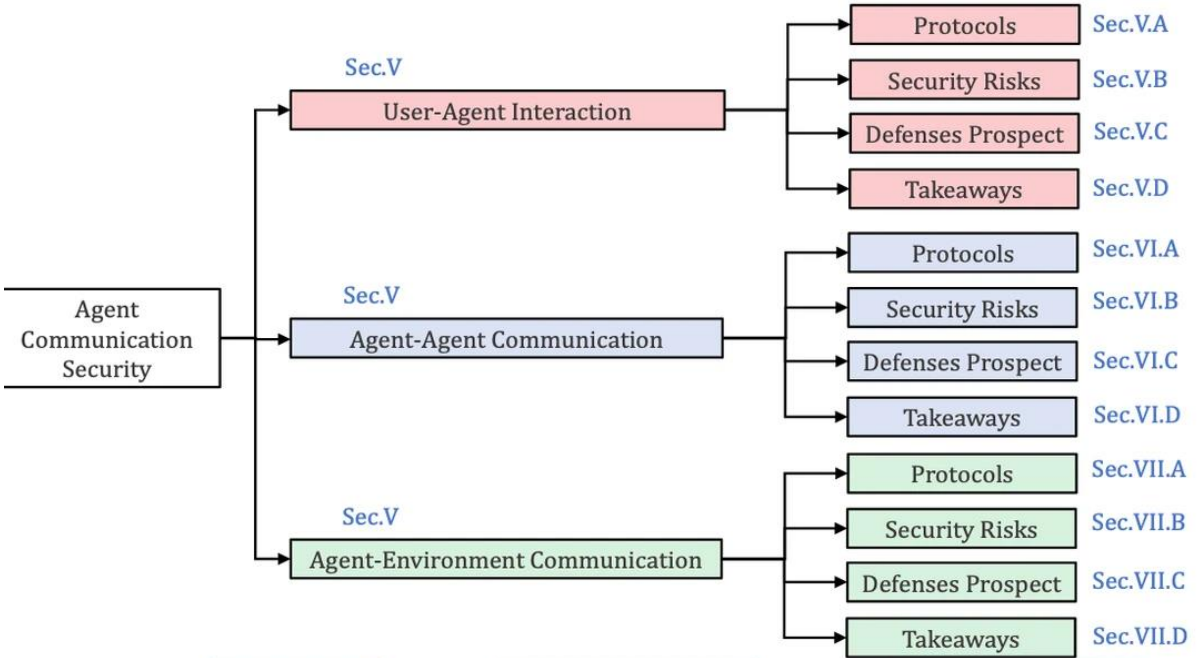
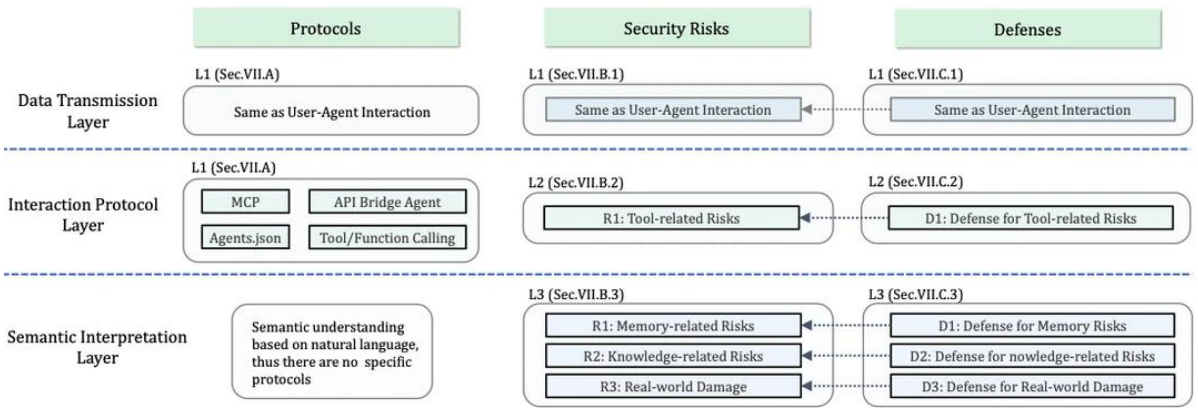


| Anomaly Detection Workflow                                                    |                                                                                                                                                             | Human Capabilities and Execution                                                 |                                                                                                                                                                                                | LLM Capabilities and Empowerment                                                 |                                                                                                                                                     |
|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>(a) Anomaly Detection</b><br><b>Tool-dependent Task</b><br>Advanced anomaly detection algorithms<br>(statistical, machine learning, deep learning, etc.) |  | <b>Telemetry Data Analysts</b><br><b>Tool Operation and Professional Skills</b><br>Debug and deploy anomaly detection algorithms,<br>set thresholds and monitor system data                    |  | <b>LLM</b><br><b>Function Calling and Tool Integration</b><br>Function Call,<br>MCP (Model Context Protocol)                                        |
|                                                                               | <b>(b) Work Condition Recognition</b><br><b>Logical Reasoning Task</b><br>Well-defined logic rule engines,<br>decision trees, etc.                          |                                                                                  | <b>Space Mission Operators</b><br><b>Logical Reasoning and Judgment</b><br>Monitor telemetry and calculate relevant indicators<br>to determine the system's work condition                     |                                                                                  | <b>LLM</b><br><b>Reasoning and Chained Thinking</b><br>Prompt Engineering, CoT (Chain-of-Thought),<br>Step-wise Prompt, Zero Shot                   |
|                                                                               | <b>(c) Anomaly Localization</b><br><b>Learning-and-approximation Task</b><br>Historical anomaly cases and<br>anomaly monitoring signals, etc.               |                                                                                  | <b>Senior Subsystem Experts</b><br><b>Experience Accumulation and Pattern Learning</b><br>Continuous research and summarize to accumulate<br>experience, analyze data to identify anomaly type |                                                                                  | <b>LLM</b><br><b>Learning and Approximation from Data</b><br>Fine-Tuning, LoRA (Low-Rank Adaptation),<br>GRPO (Group Relative Policy Optimization)  |
|                                                                               | <b>(d) Maintenance Decision-making</b><br><b>Knowledge-intensive Task</b><br>Maintenance manuals, technical<br>documents and operating procedures, etc.     |                                                                                  | <b>On-orbit Response Teams</b><br><b>Search, Summarization and Decision-making</b><br>Consult technical documents, maintenance manuals,<br>relevant procedures to formulate maintenance plans. |                                                                                  | <b>LLM</b><br><b>Knowledge Base Querying and Synthesis</b><br>RAG (Retrieval-Augmented Generation),<br>MCP (Model Context Protocol), Knowledge Base |

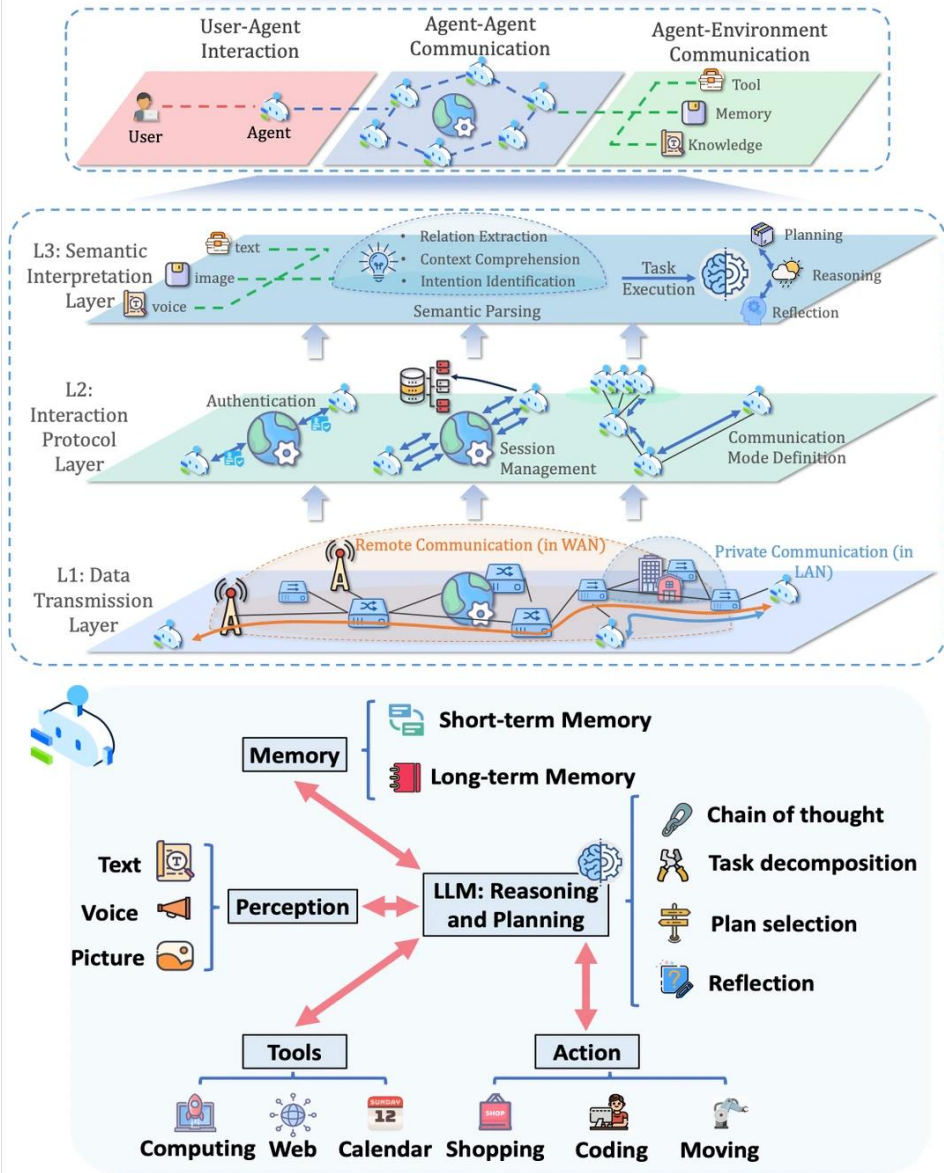
←

能力：学习

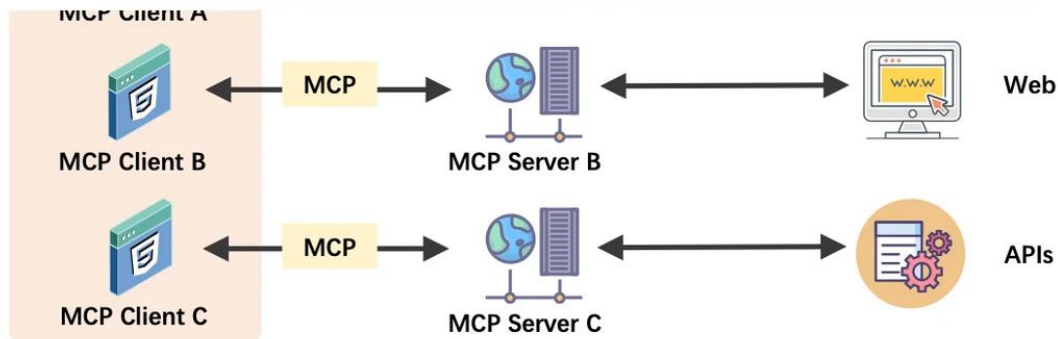
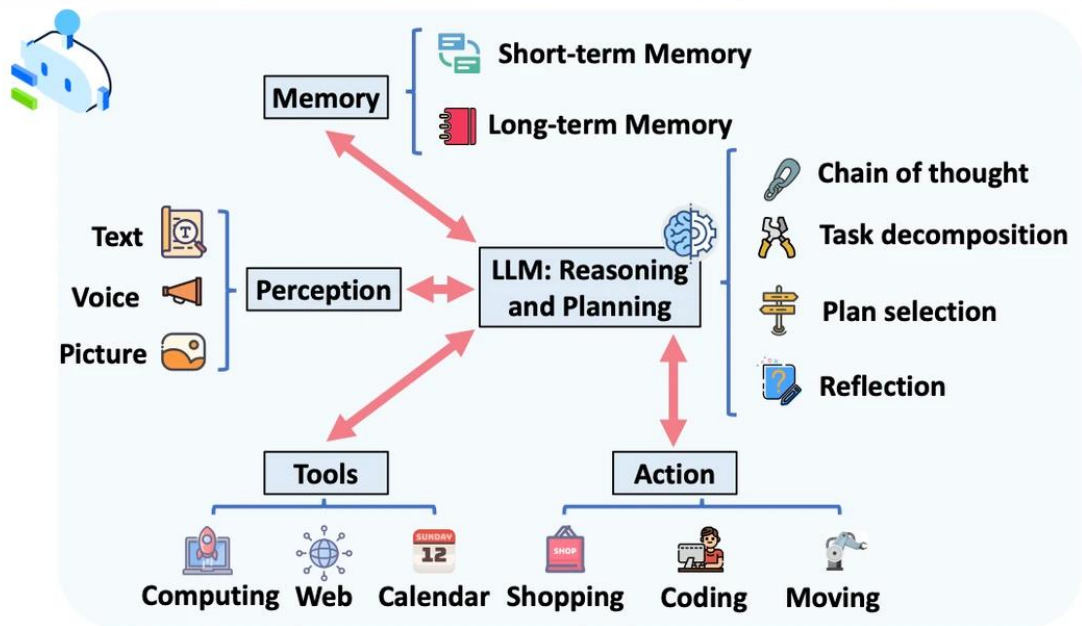
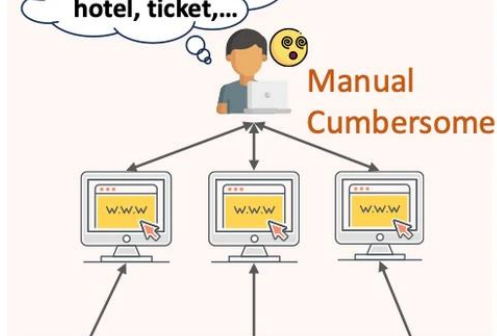
操作：调试



## Agent Communication

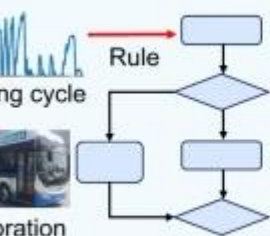




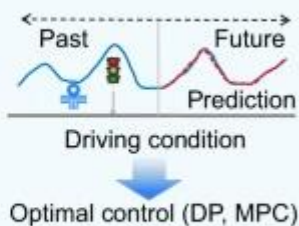




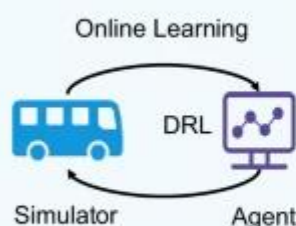
## Rule-based EMS



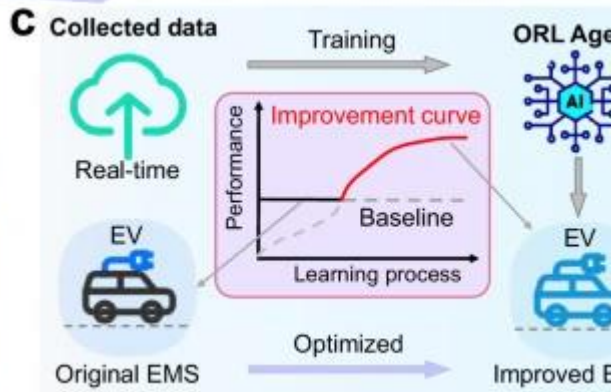
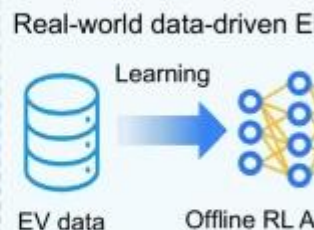
## Prediction-based

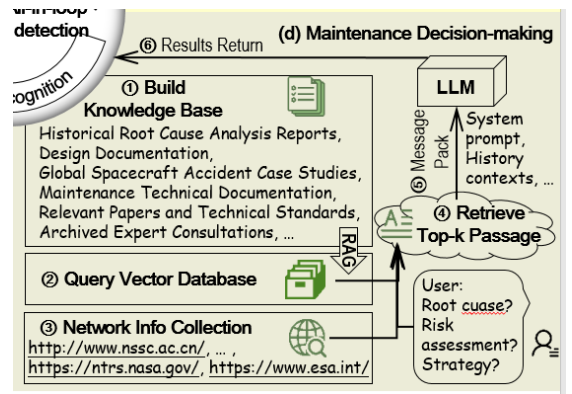


## Simulation-based



## Our approach





- 从vscode那里找到Top-K知识源头，画一个从专业知识表格文档和网页集合的图

- 左边一个红蓝表是从100个回答里面总结的
- 右边一个红蓝表是NASA的危险定级表
- 左边合理

### Background | Maintenance Data Mining

Maintenance data, documented by technicians during O&M activities, provides a comprehensive understanding of system lifelong failures and maintenance.

- System configuration;
- Failure mechanisms;
- Failure criticality;
- Maintenance actions;
- O&M insights;

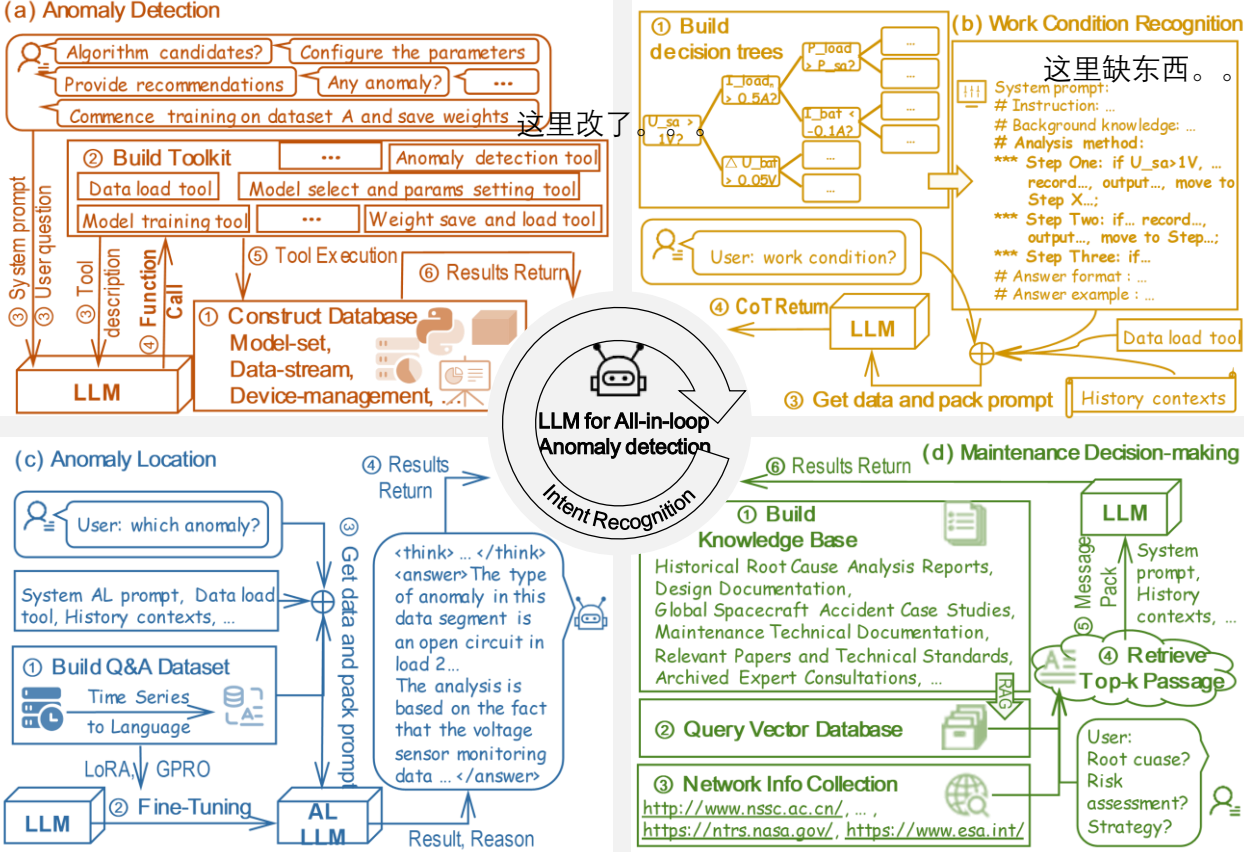
This information is instructive for subsequent O&M actions, leading to various knowledge mining and management tools:

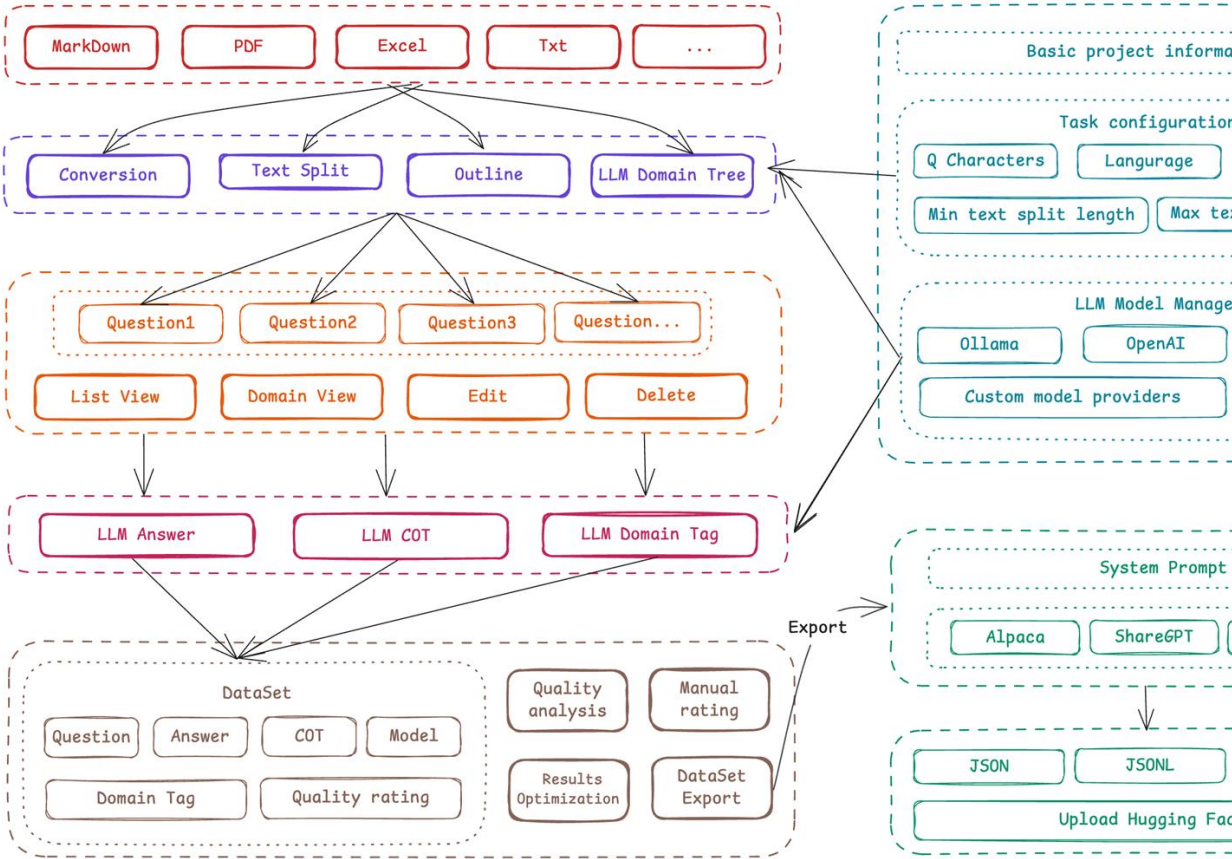
- Reliability analysis;
- FTA, FMEA, FMECA;

Traditional methods analyze maintenance data in manual. The big data environment challenges the conventional checking-based methods.

Some companies waive to analyze these data due to the prohibitive labor cost resulting from

| Maintenance Data Mining |                            |                          |                           | NASA Hazard Rating Table |    |    |    |
|-------------------------|----------------------------|--------------------------|---------------------------|--------------------------|----|----|----|
|                         |                            |                          |                           | 1                        | 2  | 3  | 4  |
| 1                       | 1.1. Configuration         | 1.2. Failure Mechanisms  | 1.3. Failure Criticality  | 5                        | 12 | 5  | 12 |
| 2                       | 2.1. System Configuration  | 2.2. Failure Mechanisms  | 2.3. Failure Criticality  | 5                        | 9  | 12 | 12 |
| 3                       | 3.1. System Configuration  | 3.2. Failure Mechanisms  | 3.3. Failure Criticality  | 5                        | 9  | 12 | 12 |
| 4                       | 4.1. System Configuration  | 4.2. Failure Mechanisms  | 4.3. Failure Criticality  | 5                        | 9  | 12 | 12 |
| 5                       | 5.1. System Configuration  | 5.2. Failure Mechanisms  | 5.3. Failure Criticality  | 5                        | 9  | 12 | 12 |
| 6                       | 6.1. System Configuration  | 6.2. Failure Mechanisms  | 6.3. Failure Criticality  | 5                        | 9  | 12 | 12 |
| 7                       | 7.1. System Configuration  | 7.2. Failure Mechanisms  | 7.3. Failure Criticality  | 5                        | 9  | 12 | 12 |
| 8                       | 8.1. System Configuration  | 8.2. Failure Mechanisms  | 8.3. Failure Criticality  | 5                        | 9  | 12 | 12 |
| 9                       | 9.1. System Configuration  | 9.2. Failure Mechanisms  | 9.3. Failure Criticality  | 5                        | 9  | 12 | 12 |
| 10                      | 10.1. System Configuration | 10.2. Failure Mechanisms | 10.3. Failure Criticality | 5                        | 9  | 12 | 12 |







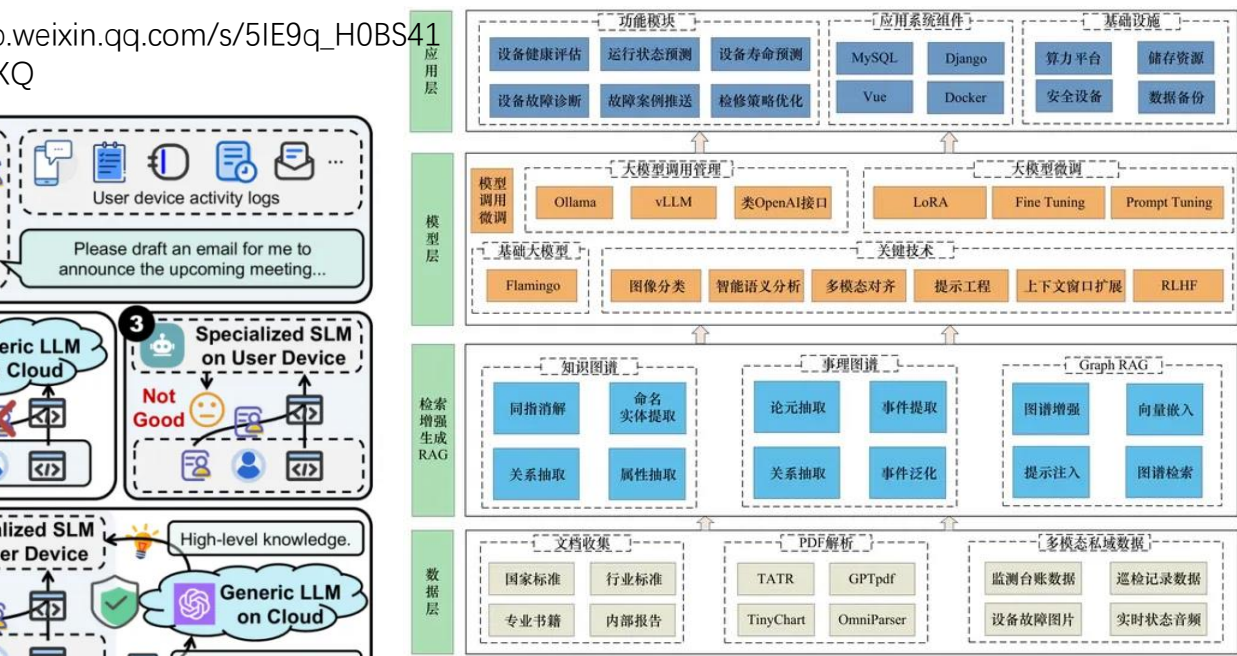
异常检测

异常检测后处理

|                                                                                                         |      |                                                                   |                                    |                                                    |                             |
|---------------------------------------------------------------------------------------------------------|------|-------------------------------------------------------------------|------------------------------------|----------------------------------------------------|-----------------------------|
| Done!<br>by<br><br>RNNs/<br>TCNs/<br>Transformers/<br>LLMs/<br>.....<br><br>Function<br>Calling、<br>MCP | 阶段   | 正常状态辨识                                                            | 异常状态定位                             | 异常检修溯源决策                                           | 其他                          |
|                                                                                                         | 任务   | 工作模式识别                                                            | 异常类型定位                             | 异常溯源、成因分析、风险评判、严重程度定级、辅助决策和检修策略推荐等                 | 知识助手、分析方案推荐、可视化代码生成、案例全流程归档 |
|                                                                                                         | 任务类型 | 对话交互型任务                                                           | 创造推理型任务                            | 专业知识依赖型任务                                          | 世界知识型任务                     |
|                                                                                                         | 硬性条件 | 工作模式识别逻辑                                                          | 一定量用于微调的数据                         | 往年归零文档、设计文档、世界航天器事故案例收集、检修策略技术文档、相关论文及技术标准、专家咨询留档等 | 满血版API调用                    |
|                                                                                                         | 技术手段 | CoT、Prompt工程、zero shot、one shot或few shot嵌入prompt、Step-wise prompt | 预训练微调、LoRA、GPRO、时序数据输入适配、时频特征提取与构造 | RAG、MAP、知识库编撰、知识图谱                                 | API调用及部署                    |

况复杂，所以通过世界知识生成预处理代码，用到的技术就是copilient和client。。。 如果你的AI只需简单查数据，Function Calling够用；但要玩转企业级自动化？MCP才是

o.weixin.qq.com/s/5IE9q\_H0BS41  
KQ



structure diagram. 1. Context-aware instructional examples. LLMs excel in context awareness but pose privacy risks. SLMs (Small Language Models) prioritize privacy but have lower performance. LLMs and SLMs enhance privacy protection and improve



## 代码里面有所更新：代码里的+++回答样例

**## 回答示例：**同时，为你提供一些正确回答的示例可供你参考：

“<经典样本示例1>”、“<经典样本示例2>”、“<经典样本示例3>” .....

**## 注意：**1.当你按我提供的步骤进行分析时，若在同一数据段中同时出现“大于”和“小于”的情况，则说明该时段可能发生了工况切换。请你将该段划分为“切换前”和“切换后”两个阶段（若发生多次切换，则依次划分为多个阶段），分别在每个阶段依据我给出的步骤进行分析，并为每个阶段输出独立的回答。2.虽然我的异常检测算法认定该数据是正常状态，但若你发现不符合正常工作模式的特征，仍需报警。

**## 数据：**请分析，各传感器这段时间内所采集到的数据如下：<传感器1>：<x>,<x>,<x>,...;<传感器2>：<x>,<x>,<x>,...;<传感器3>：<x>,<x>,<x>,...;<传感器4>：<x>,<x>,<x>,...

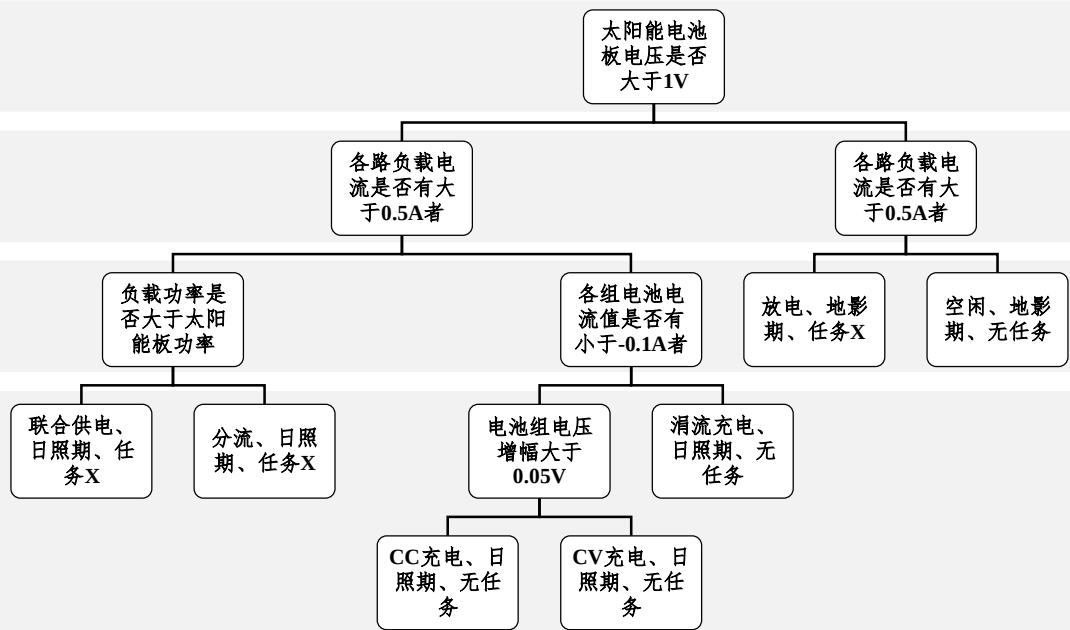


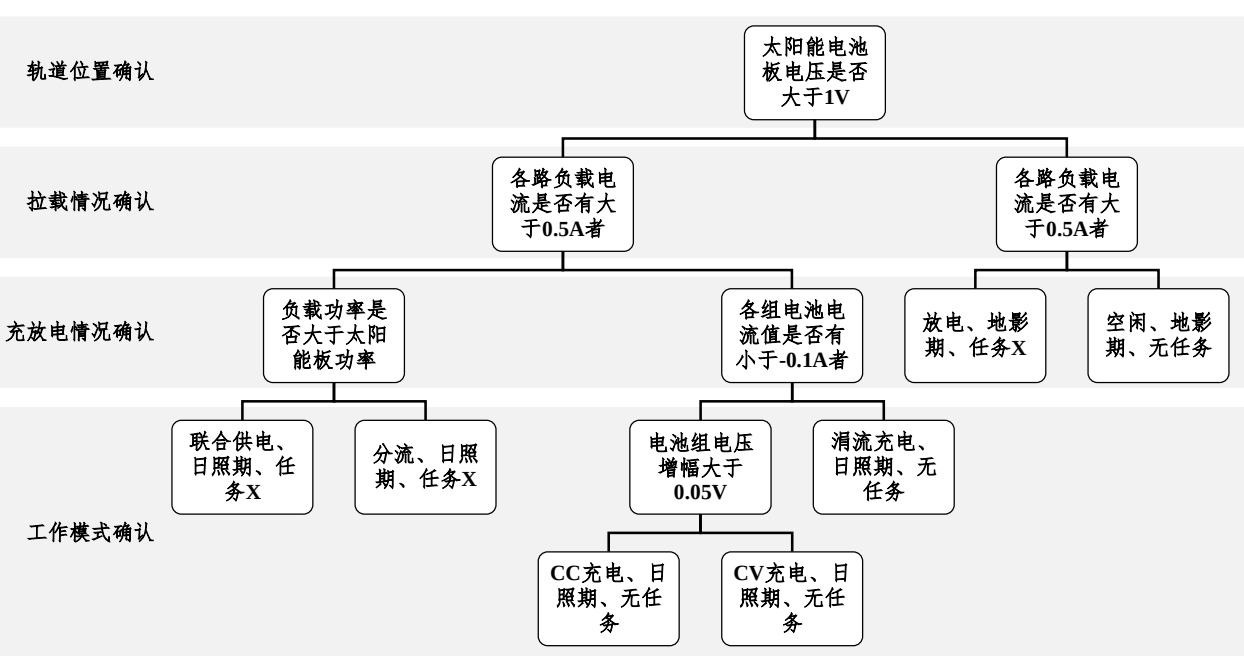
轨道位置确认

拉载情况确认

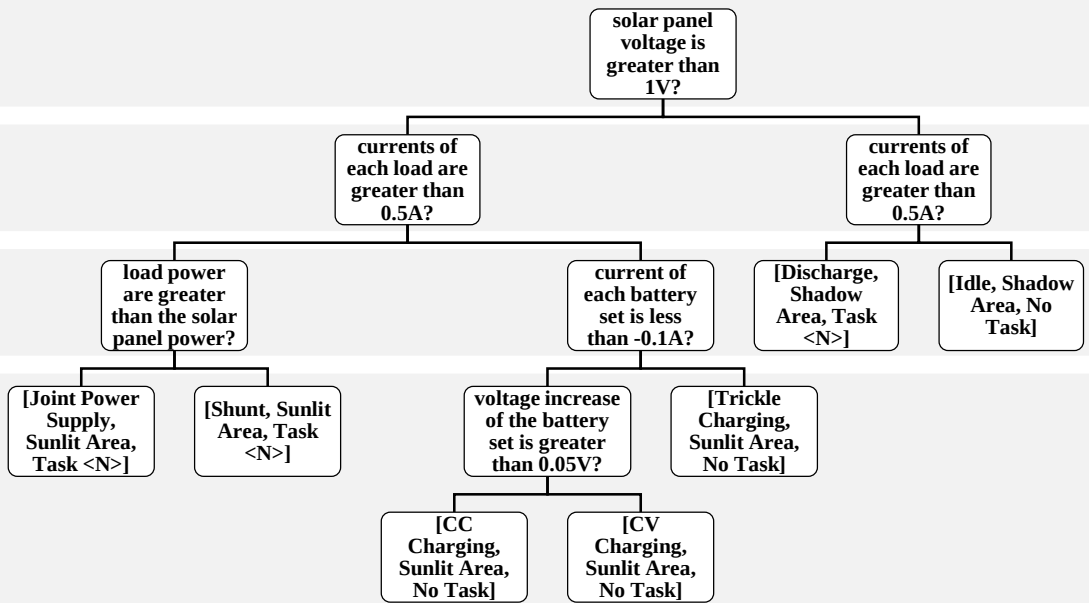
充放电情况确认

工作模式确认





这个是导向CC、CV充电，而不是导向快充和涓流充电



这个是导向CC、CV充电，而不是导向快充和涓流充电

**## 任务：**我正在进行异常检测的后处理，现需要进行正常状态的信息确认，请你根据我提供的信息为我分析该设备现在的工作模式等状态。

**## 相关知识：**已知该段数据采集于一个航天器电源系统，其由太阳能电池板、3组蓄电池组、3路负载、充电控制器BCR、放电调节器BDR、分流调节器SR、母线和功率分配模块PDM等子系统或部件组成。

**## 数据：**在过去<N>个时间步所采集到的数据如下：<传感器1>：<x>,<x>,<x>,...; <传感器2>：<x>,<x>,<x>,...; <传感器3>：<x>,<x>,<x>,...; <传感器4>：<x>,<x>,<x>,...

**## 分析方法：**请根据以下步骤进行分析：

\*\*\*步骤1：确认太阳能电池板电压是否大于1V？若是，记录为日照期，转入步骤2；若否，记录为地影期，转入步骤6；

\*\*\*步骤2：确认各路负载电流是否大于0.5A？若是，第<N>路负载<N>电流大于0.5A则记录为正在执行任务<N>，转入步骤3；若否，记录为无任务，转入步骤4；

\*\*\*步骤3：确认负载<N>功率是否大于太阳能电池板功率？若是，输出工作状态为【联合供电、日照期、任务<N>】；若否，输出工作状态为【分流、日照期、任务<N>】，转入步骤7；

\*\*\*步骤4：确认各组电池电流值是否小于-0.1A？若是，转入步骤5；若否，记录为涓流充电，输出工作状态为【涓流充电、日照期、无任务】，转入步骤7；

\*\*\*步骤5：确认电池组电压增幅是否大于0.05V？若是，记录为CC充电，输出工作状态为【CC充电、日照期、无任务】；若否，记录为CV充电，输出工作状态为【CV充电、日照期、无任务】，转入步骤7；

\*\*\*步骤6：确认各路负载电流是否大于0.5A？若是，第<N>路负载<N>电流大于0.5A则记录为正在执行任务<N>，输出工作状态为【放电、地影期、任务<N>】；若否，记录为无任务，输出工作状态为【空闲、地影期、无任务】，转入步骤7；

\*\*\*步骤7：根据所输出工作状态检查各传感器观测值，若联合供电，<传感器1>工作范围是否在<XXX>？<传感器2>工作范围是否在<XXX>？..... <传感器X>工作范围是否在<XXX>？

**## 回答格式：**你回答问题的格式必须遵照如下模板，替换填写[]和<>中的内容：

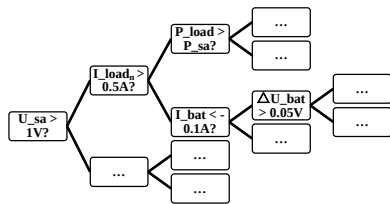
“步骤1：确认[确认内容]，[确认过程推理]，答案为[是/否]，记录为[地影期/转入其他步骤]；步骤2：确认[确认内容]，[确认过程推理]，答案为[是/否]，记录为[正在执行任务<N>/无任务/转入其他步骤]；步骤3：确认[确认内容]，[确认过程推理]，答案为[是/否]，记录为[.../转入其他步骤]；步骤4：确认[确认内容]，[确认过程推理]，答案为[是/否]，记录为[涓流充电/转入其他步骤]；步骤5：确认[确认内容]，[确认过程推理]，答案为[是/否]，记录为[CC充电/CV充电/转入其他步骤]；步骤6：确认[确认内容]，[确认过程推理]，答案为[是/否]，记录为[正在执行任务<N>/无任务/转入其他步骤]；步骤7：根据所输出工作状态检查各传感器观测值：[检查内容]，[检查过程推理]，记录为[检查正常/检查不通过，结果不可信]；综上，该航天器目前的工作状态为：[输出工作状态]。”

**## 回答示例：**同时，为你提供一些正确回答的示例可供你参考：

“<经典样本示例1>”、“<经典样本示例2>”、“<经典样本示例3>” .....

**## 注意：**虽然我的异常检测算法认定该数据是正常状态，但若你发现不符合正常工作模式的特征，仍需报警。







**## 背景：**我们正在进行异常检测的后处理，已知运维对象为航天器电源系统，其由太阳能电池板（负责将光能转为电能）、3组蓄电池组（负责储存电能）、3路负载（负责消耗电能和执行任务）、充电调节器BCR（负责调节电池充电、放电、分流）、放电调节器BDR、分流调节器SR、母线（负责传输电能）和功率分配模块PDM（负责分配电能到各路负载）等子系统或部件组成。

**## 任务：**我们已经定位到当前发生的是<太阳能电池板部分元件或支路开路/太阳能电池板部分元件或支路短路/太阳能电池板元件老化/BCR开路/BCR短路/BCR功率损耗增加/电池组老化/电池组开路/母线开路/母线短路/母线绝缘击穿/第<N>路PDM开路或短路//第<N>路负载开路.....>异常，表现为<.....>。请你通过查询、分析我的知识库文件夹，为我完成异常溯源、成因分析、风险评判、严重程度定级、辅助决策和检修策略推荐等任务。

**## 回答格式：**你回答问题的格式最好能遵照如下模板，替换填写补充[]和<>和.....中的内容，可以进行适当的改写和拓展：

“步骤1—异常归因：查询往年归零文档、相关论文、技术报告与设计文档等，总结可能导致[当前异常类型]发生的原因有：（1）[高能粒子撞击/帆板卡死/.....]，即[相关解释、分析等]、（2）[可能的原因]，即[相关解释、分析等]、（3）.....；

步骤2—风险评判：查询往年航天器电源系统类似异常案例，相似的情况有：<N>年[xxx]国[xxx]航天器发生了[xxx]，导致[xxx]后果].....<N>年[xxx]国[xxx]航天器发生了[xxx]等，导致[xxx]后果]。同时，分析相关论文、归零报告、技术文件，可总结该类型异常可能导致的后果有：[后果一xxx、后果二xxx、.....]，因此，建议将该异常定档为[灾难级/严重/一般/轻微]；

步骤3—检修策略：查询检修策略文件夹、专家咨询留档、相关论文和技术报告等，总结该异常可能的检修策略包括：[系统重启.....]、[供配电策略切换.....]、[启动冗余备份单元.....]、.....；

步骤4：以上回答参考文献位置：[知识库文件位置/网页链接/.....]。”

**步骤1—异常归因：**查询往年归零文档、相关论文、技术报告与设计文档等，总结可能导致【太阳能电池板部分元件或支路开路】异常发生的原因有：

(1) **Picoblade接插件松脱失效**<sup>[^1]</sup>，即帆板背面的接插件因长期承受-60℃~90℃的极端温度交变导致老化碎裂，在卫星姿态调整时因机械振动发生松脱，造成支路断路；

(2) **单粒子事件引发元器件损坏**<sup>[^2]</sup>，即高能质子或重离子撞击导致光伏组件断路失效，但概率较低且可通过防护设计规避；

(3) **帆板展开机构卡滞引发局部应力集中**<sup>[^3]</sup>，即帆板展开未完全到位导致部分电池片连接线缆长期处于弯曲应力状态，最终引发金属疲劳断裂。

**步骤2—风险评判：**查询航天器电源系统类似异常案例，相似的情况有：

- 2008年 **美国Galaxy 26卫星**<sup>[^4]</sup>因微陨石撞击导致太阳能电池阵开路，损失50%发电能力，最终提前退役；

- 2010年 **俄罗斯斯特赫-1通信卫星**<sup>[^5]</sup>因太阳翼展开故障引发开路，致使卫星运行6个月后失效。

同时，分析相关技术文档可知，该异常可能导致**整星供能不足、蓄电池过放失效、关键载荷中断等灾难性后果**<sup>[^6]</sup>，建议定级为**严重级（需立即干预但未完全丧失功能）**。

**步骤3—检修策略：**查询检修策略文件夹、专家咨询留档、相关论文和技术报告等，总结可能的检修策略包括：

(1) **自主供电策略切换：**启用蓄电池冗余模块并调整V-T充电曲线<sup>[^7]</sup>，优先保障核心系统供电；

(2) **硬件冗余激活：**若存在备份帆板支路，通过指令切换至冗余电路<sup>[^8]</sup>；

(3) **在轨维护操作：**通过机械臂（若有）检查接插件状态并尝试物理复接<sup>[^9]</sup>；

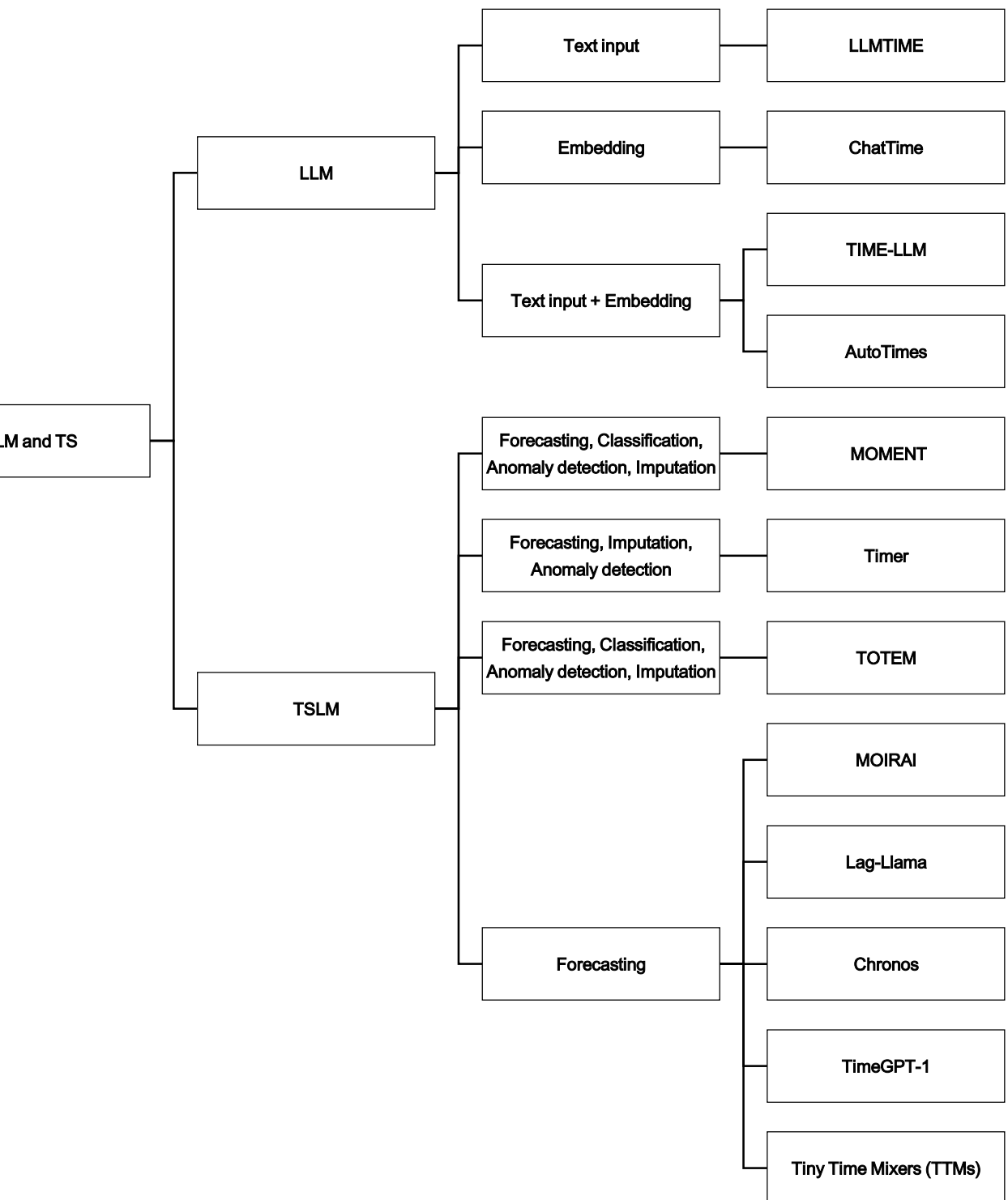
(4) **帆板角度动态优化：**调整卫星姿态减小太阳光照角 $\theta$ ，最大限度提升可用支路的输出功率<sup>[^10]</sup>。

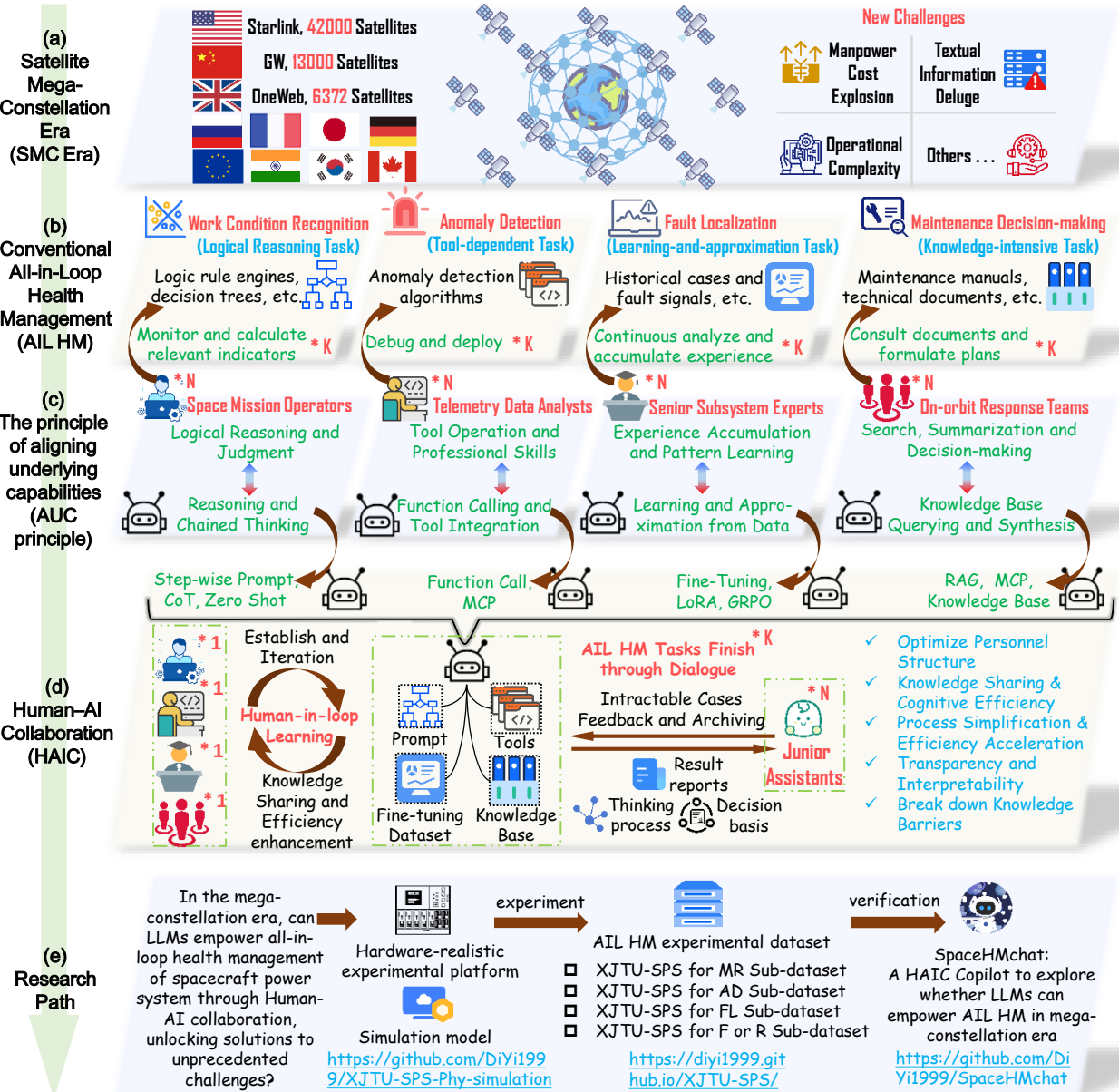
**步骤4—以上回答参考文献位置：**

[^1]: E:\OneDrive\DATA\Our\_Exp\_Data\XJTU-SPS Dataset\Knowledge Base\归零文档\EPS在轨不充电问题归零报告.docx (id=2)

.....（文档众多此处省略）







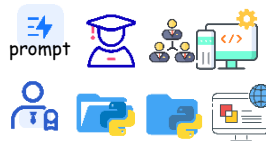
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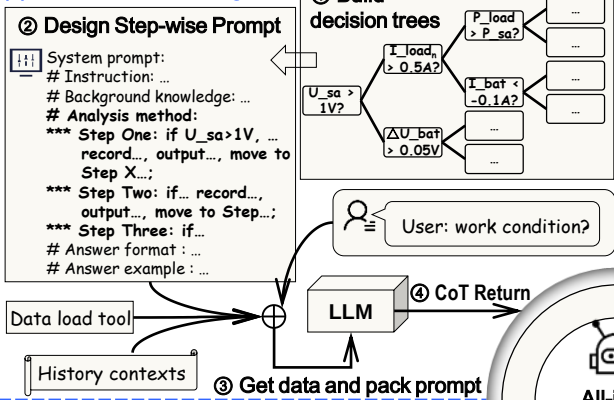
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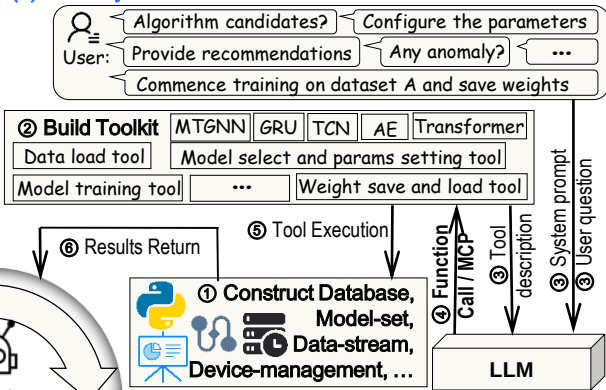


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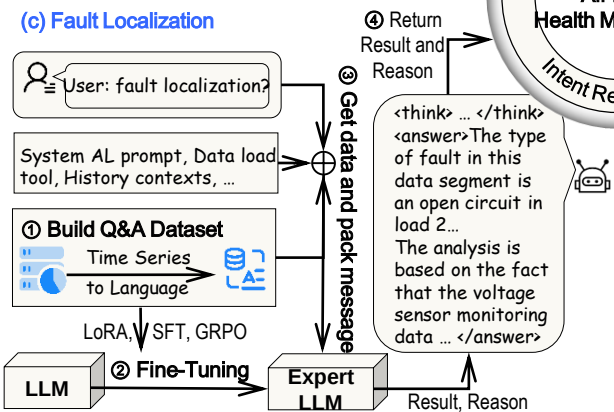
## (a) Work Condition Recognition



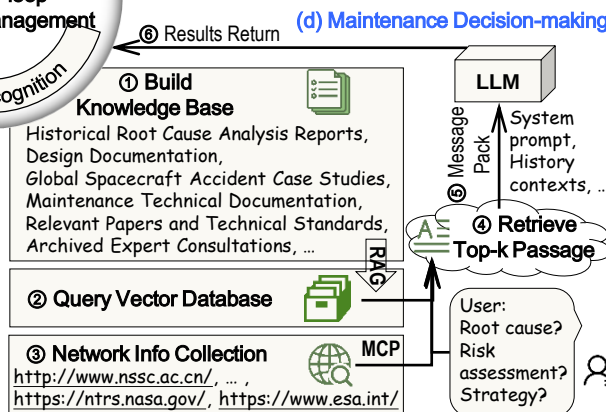
## (b) Anomaly Detection



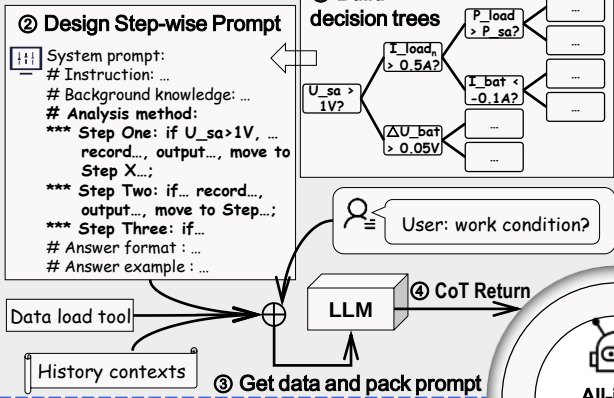
## (c) Fault Localization



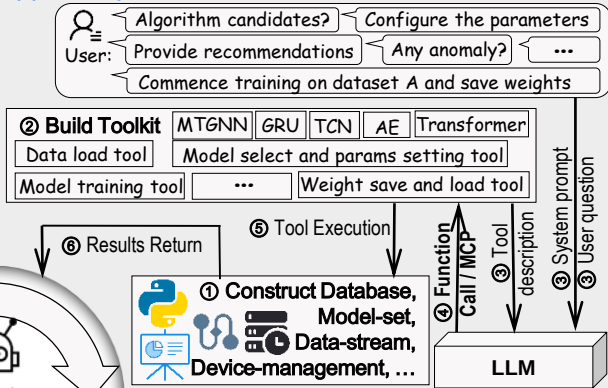
## (d) Maintenance Decision-making



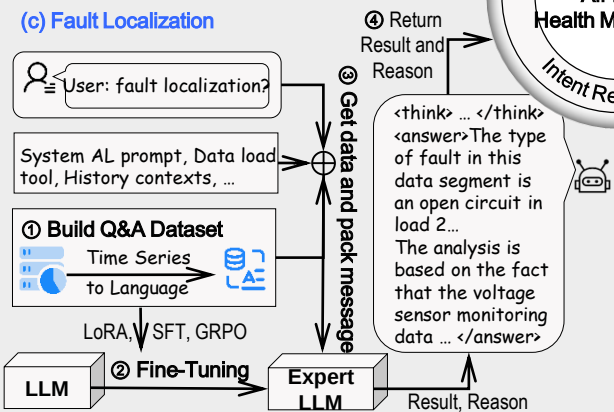
## (a) Work Condition Recognition



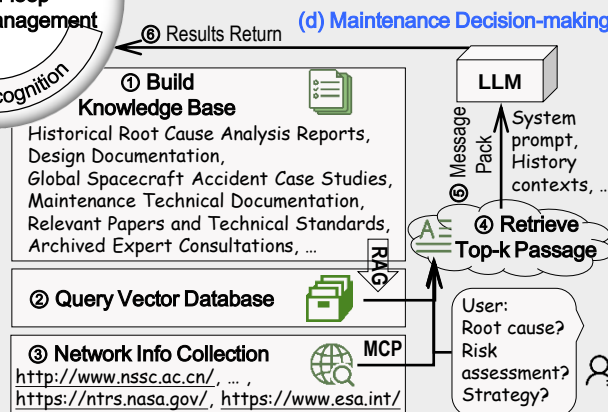
## (b) Anomaly Detection



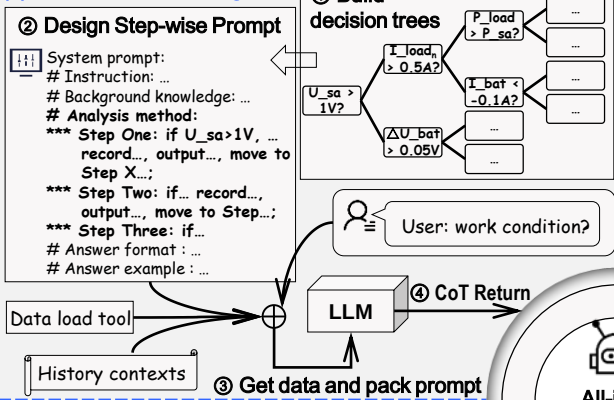
## (c) Fault Localization



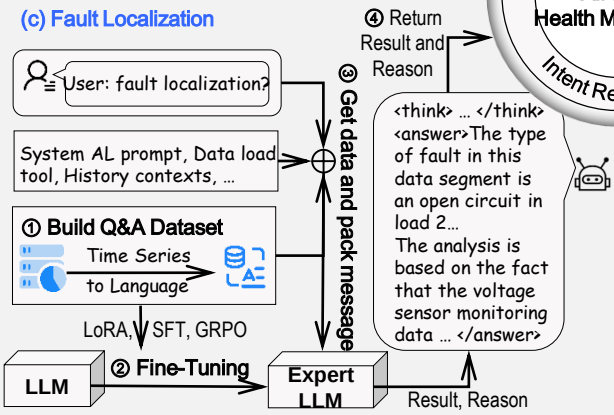
## (d) Maintenance Decision-making



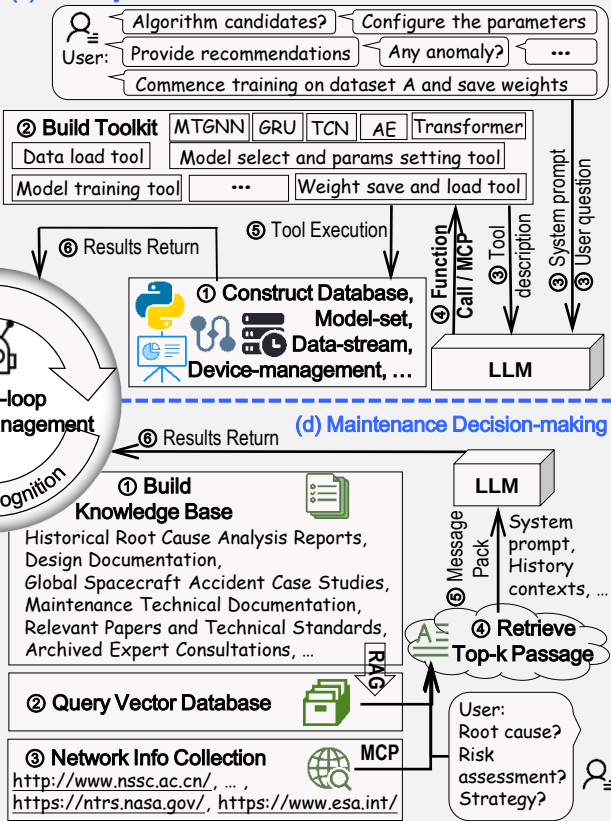
### (a) Work Condition Recognition



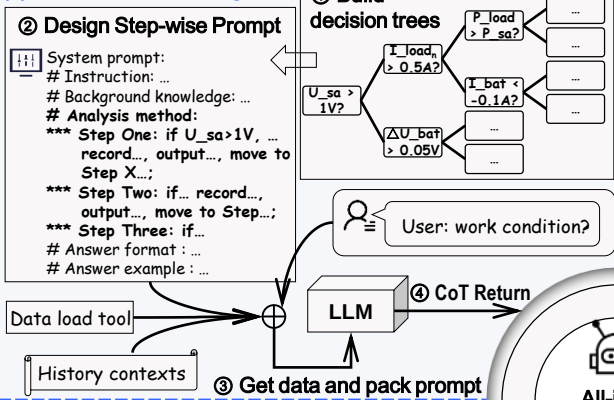
### (c) Fault Localization



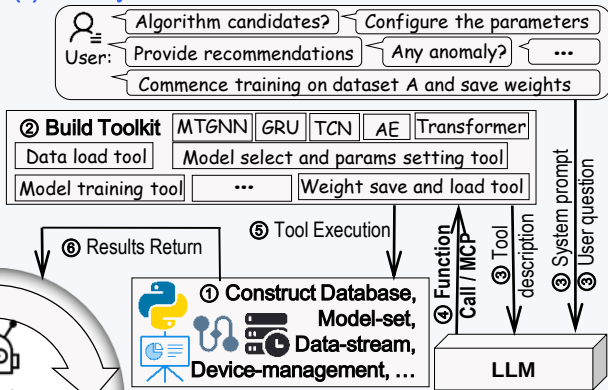
### (b) Anomaly Detection



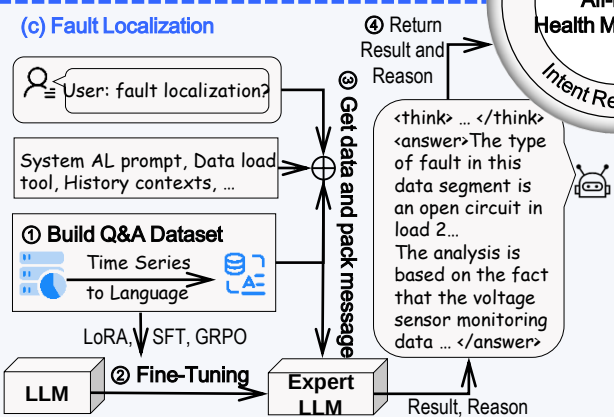
## (a) Work Condition Recognition



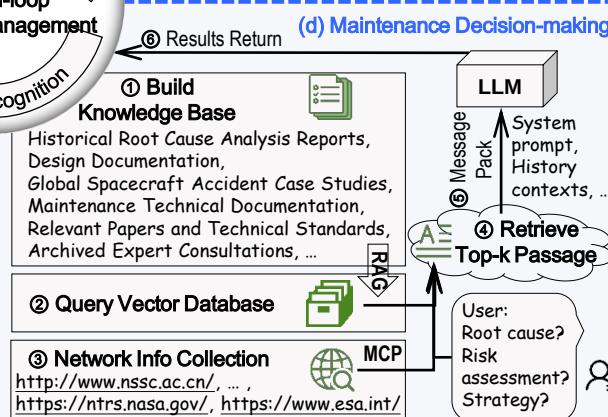
## (b) Anomaly Detection



## (c) Fault Localization

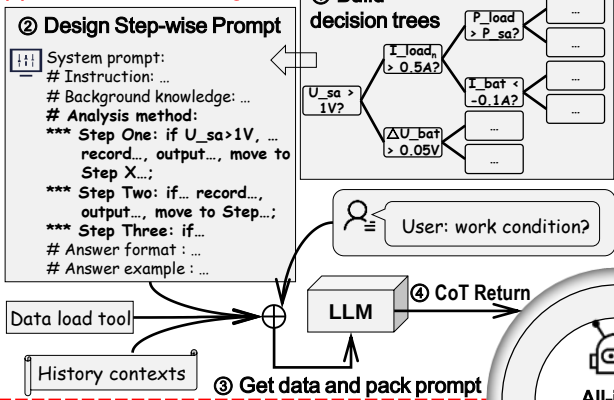


## (d) Maintenance Decision-making

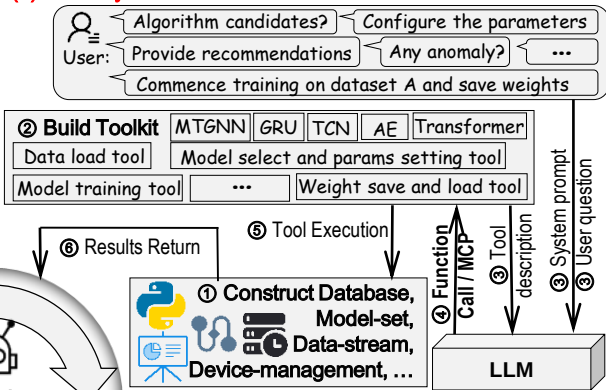




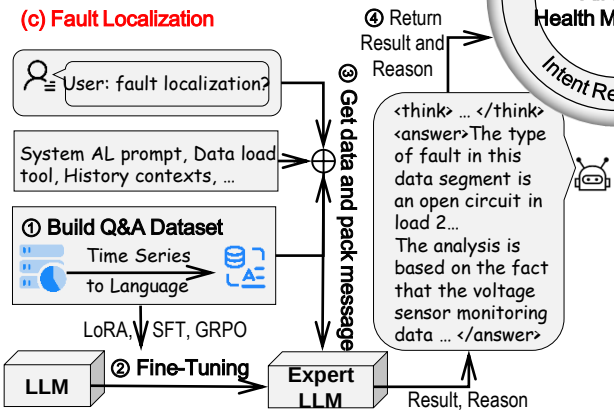
### (a) Work Condition Recognition



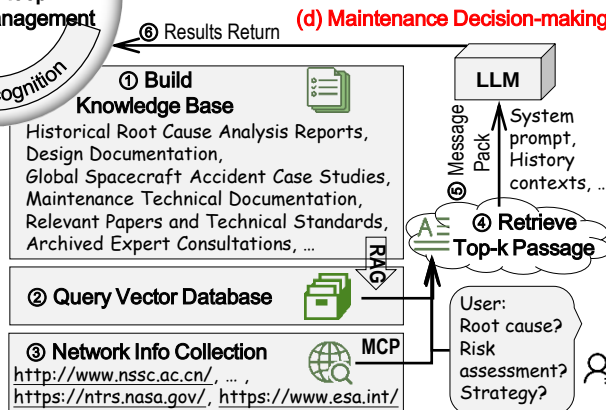
### (b) Anomaly Detection



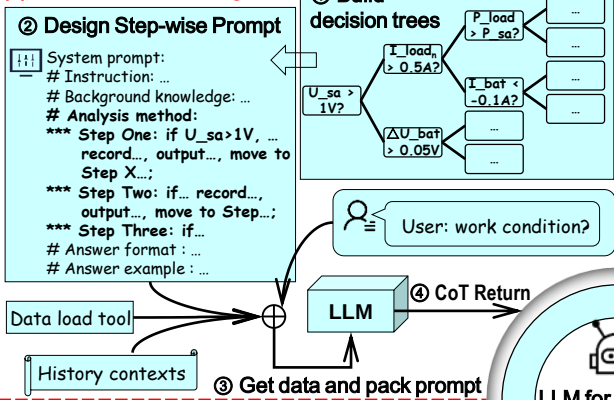
### (c) Fault Localization



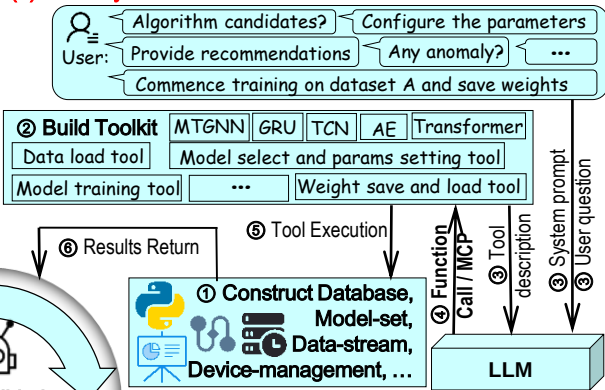
#### (d) Maintenance Decision-making



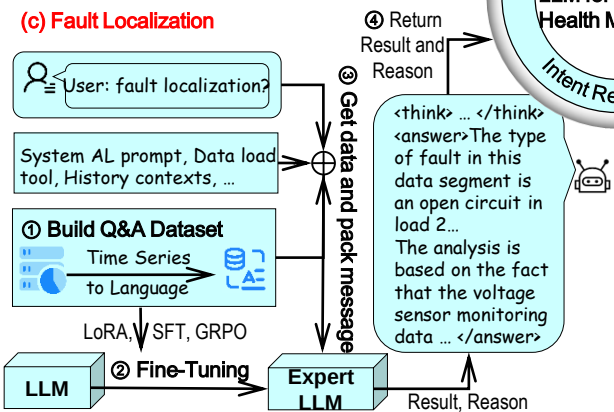
## (a) Work Condition Recognition



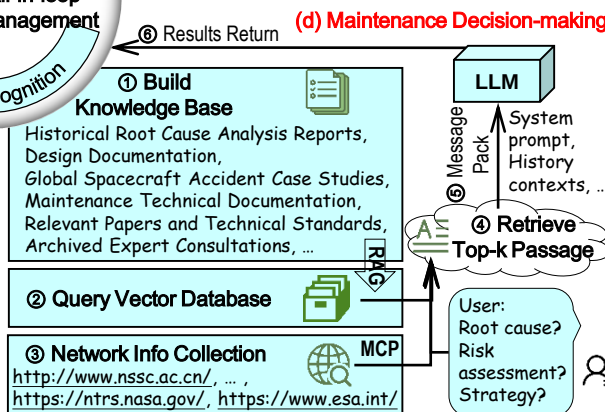
## (b) Anomaly Detection

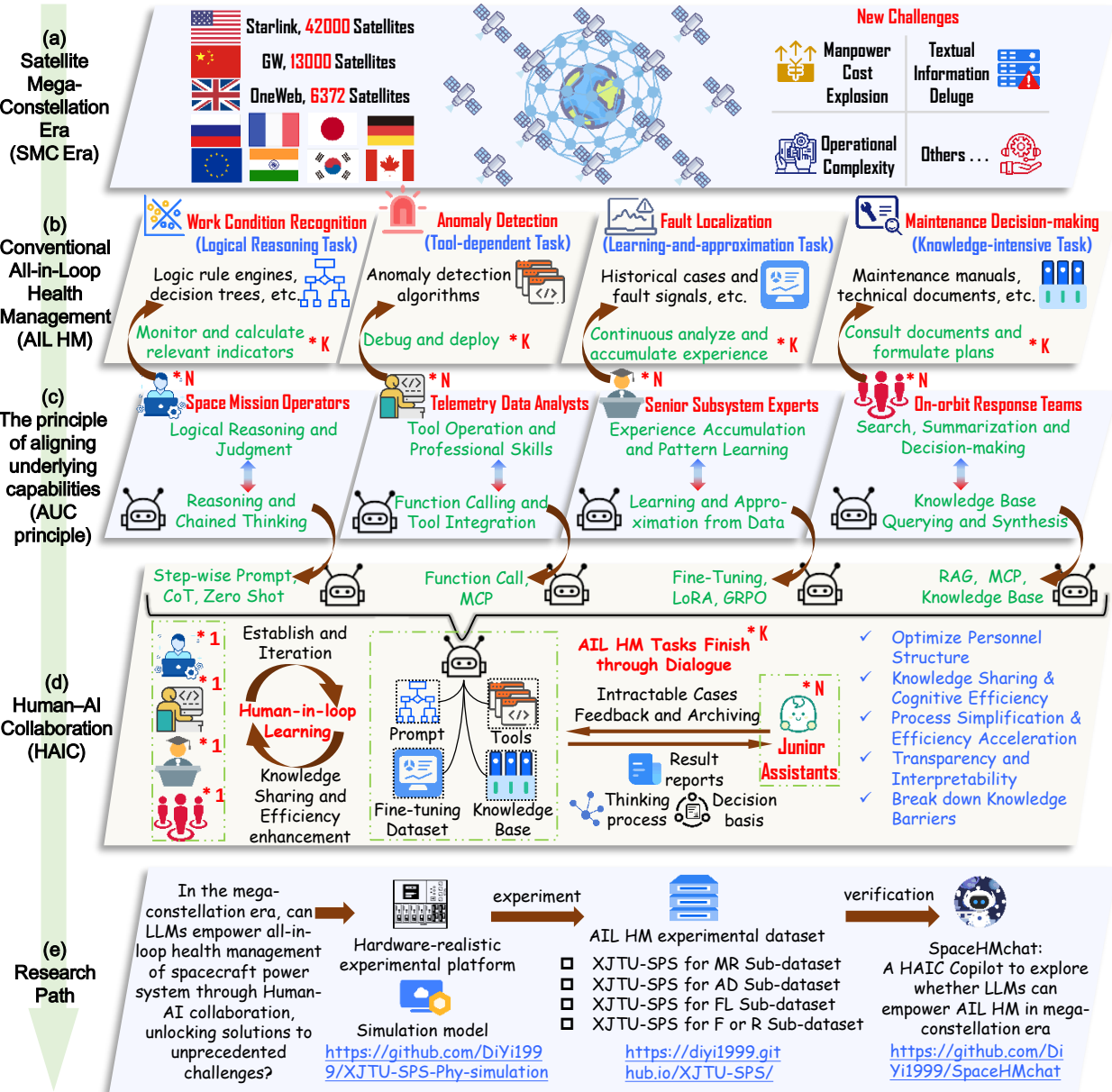


## (c) Fault Localization



## (d) Maintenance Decision-making





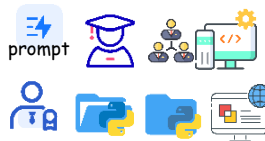
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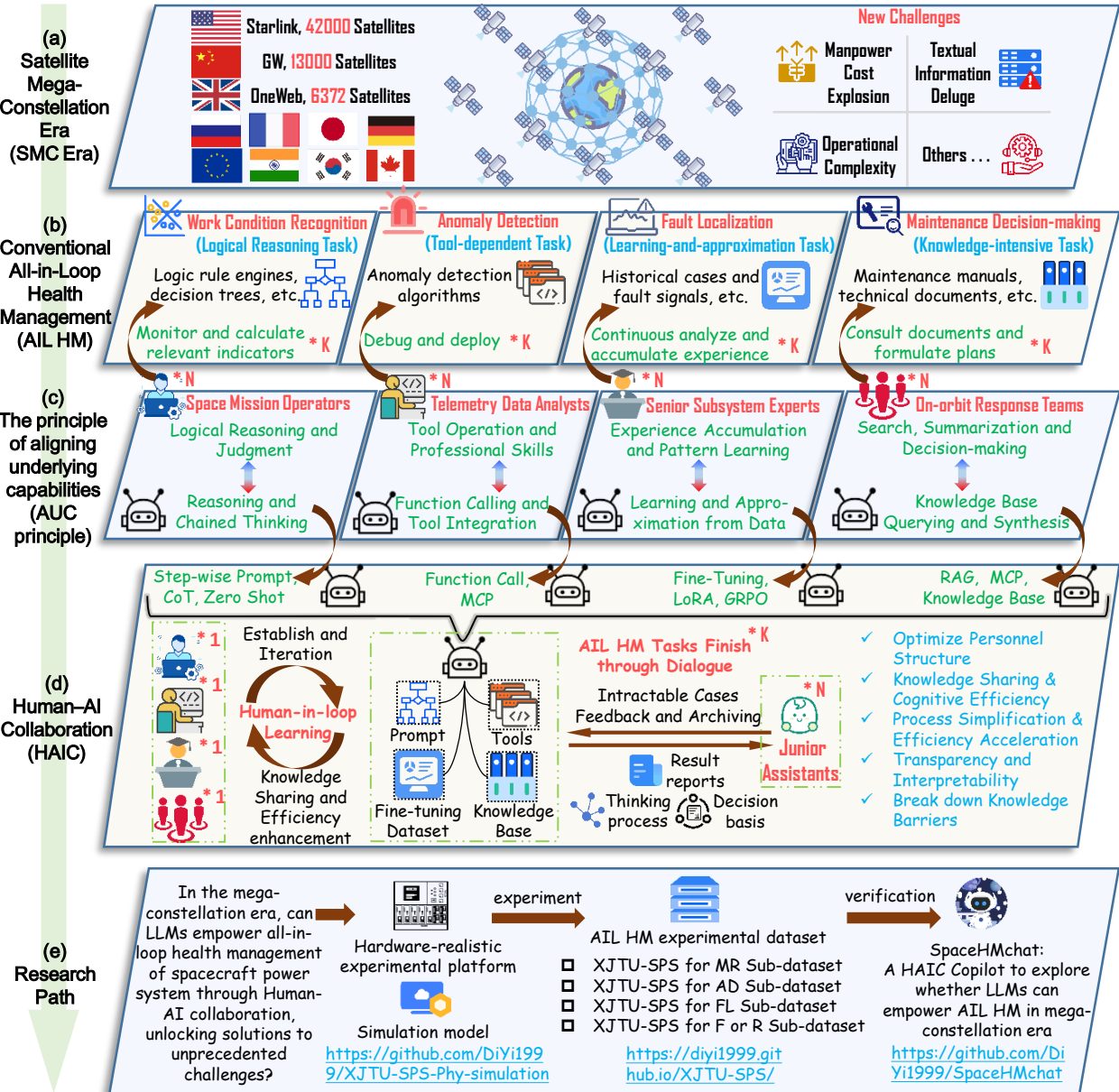
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unprecedented challenges in the mega-constellation era !

This work explores whether large language models can empower spacecraft health management in the mega-constellation era by establishing an effective Human-AI collaboration paradigm.



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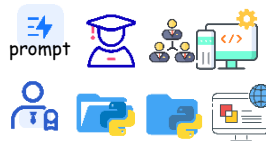
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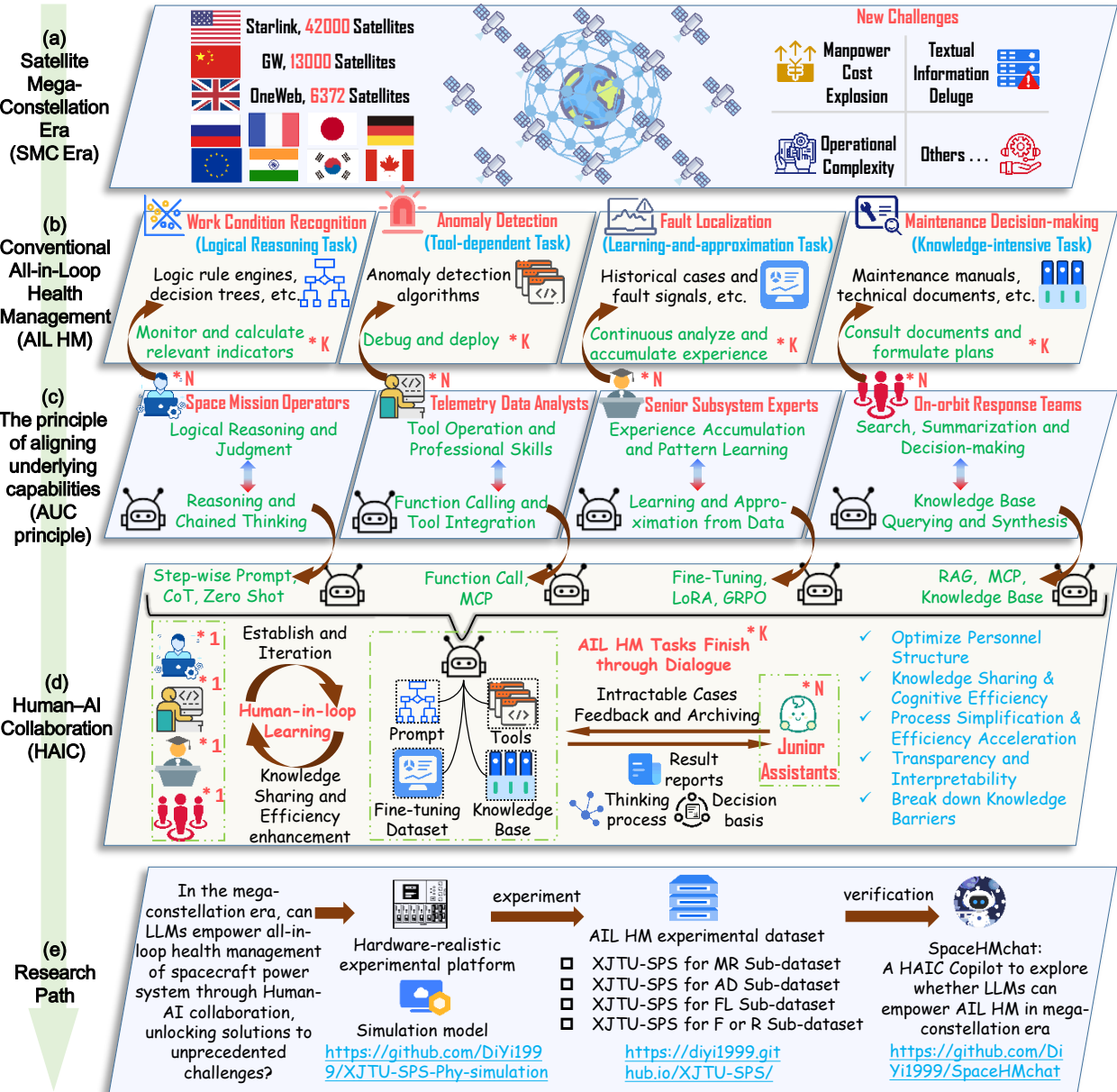
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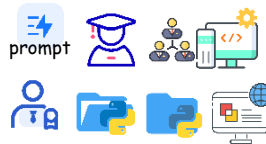
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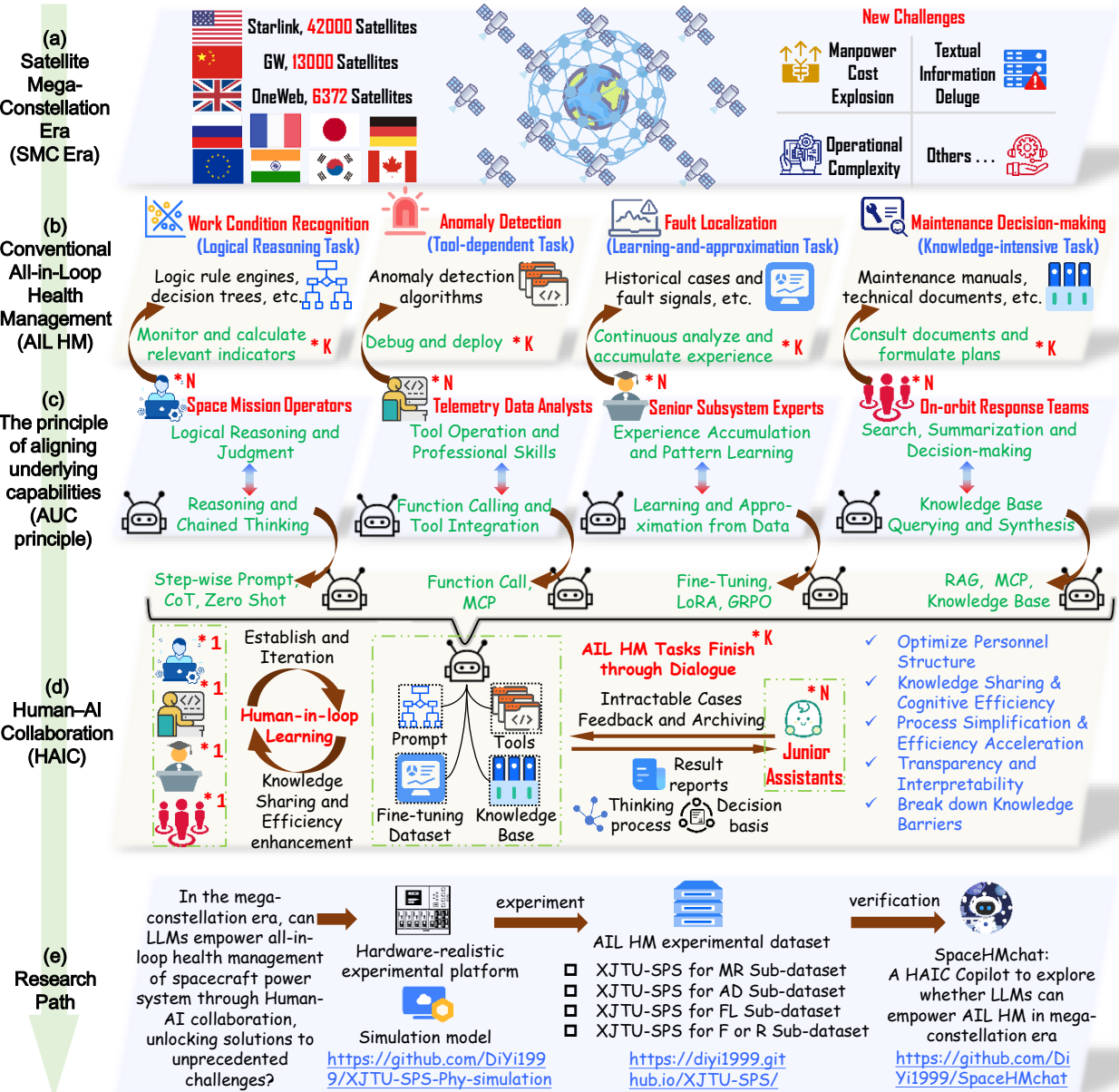
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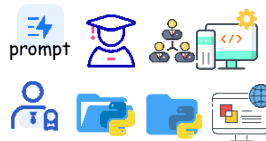
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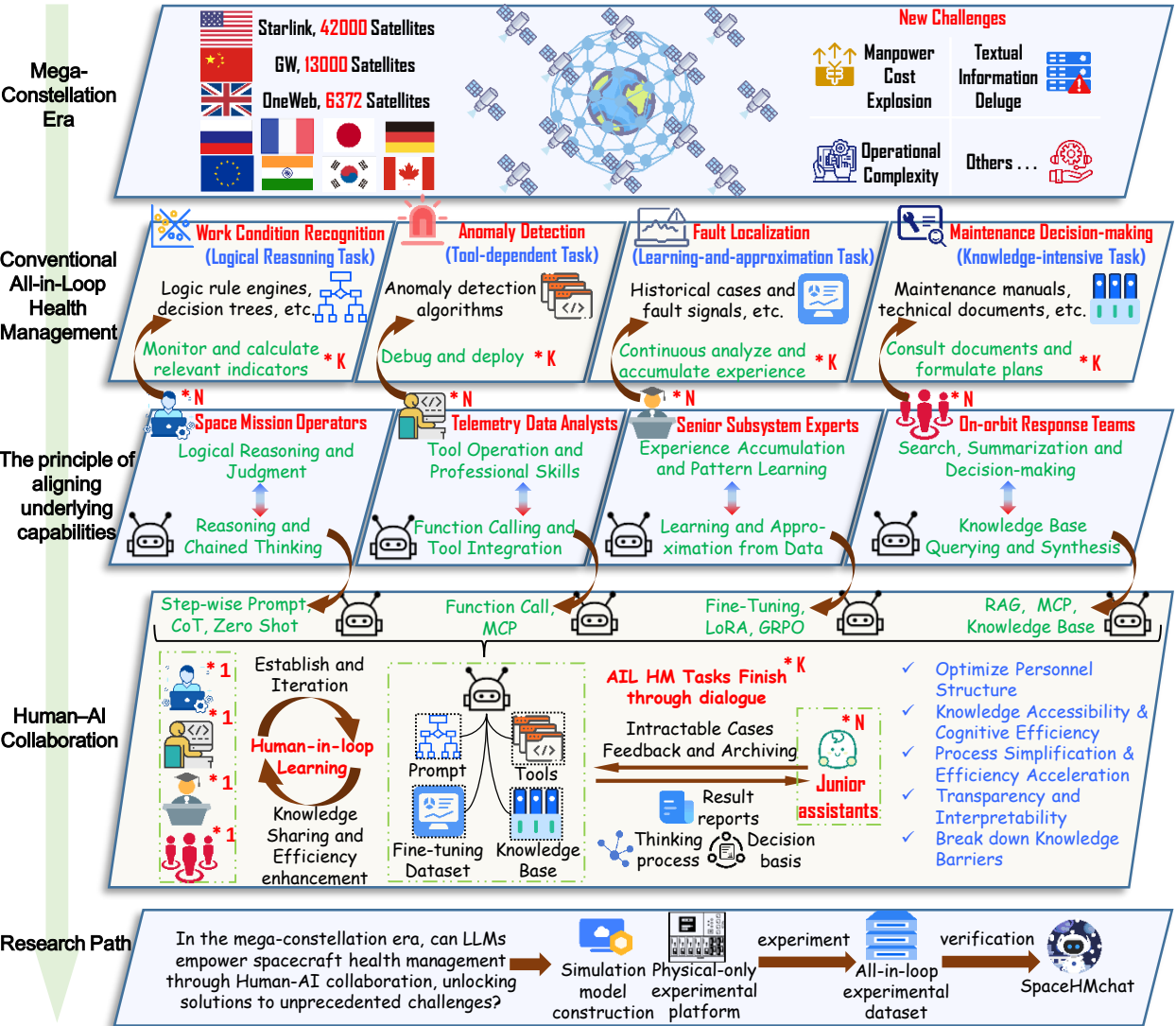
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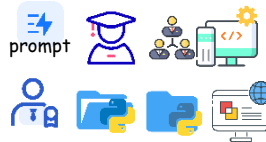
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## ② Design Step-wise Prompt

